

ZHIZHGIN, Russian Federation

WMO: 224380

Lat: 65.200N

Lon: 36.817E

Elev: 17

StdP: 101.12

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.5	-16.6	-20.6	0.6	-18.1	-18.7	0.7	-16.3	18.4	-8.4	17.1	-10.4	4.7	110	0.853

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.7	20.3	16.4	18.6	15.4	17.2	14.5	17.0	19.4	15.9	18.1	14.8	16.8	4.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.0	11.4	18.1	14.8	10.5	17.3	13.7	9.8	16.1	47.9	19.5	44.5	18.1	41.6	16.9	21.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.9	11.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-7.6	-6.5	-4.8	-0.8	4.0	9.3	12.9	12.5	9.4	4.0	-1.2	-3.5
	DBStd	7.92	4.98	4.54	4.31	3.58	3.73	3.62	3.10	2.50	2.27	2.64	2.86	3.44
	HDD10.0	3025	544	462	458	326	189	56	8	5	38	186	336	418
	HDD18.3	5831	803	695	717	576	443	271	169	182	269	444	586	676
	CDD10.0	238	0	0	0	0	3	36	99	82	18	0	0	0
	CDD18.3	3	0	0	0	0	0	1	2	1	0	0	0	0
	CDH23.3	5	0	0	0	0	0	2	3	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	5.9	7.1	6.3	5.9	5.2	5.3	5.0	4.7	4.9	5.3	6.5	6.9	7.3
Precipitation	PrecAvg	371	18	13	16	19	27	37	39	55	47	49	35	18
	PrecMax	556	33	30	34	51	106	149	91	138	96	102	73	33
	PrecMin	252	5	1	3	1	2	5	2	18	15	2	13	2
	PrecStd	64	8	7	9	12	21	26	24	28	22	21	16	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.0	1.4	3.3	9.2	16.5	22.2	23.1	21.0	16.3	10.6	5.5	2.2
		MCWB	0.5	0.7	1.4	5.4	10.5	16.7	17.4	17.4	13.7	9.7	4.6	1.3
	2%	DB	0.3	0.8	1.7	6.1	13.7	18.7	20.8	19.2	14.3	9.4	4.6	1.4
		MCWB	-0.2	0.2	0.4	3.6	9.8	14.2	16.9	16.2	12.3	8.5	4.0	0.7
	5%	DB	-0.3	0.2	1.1	4.3	11.6	16.8	19.2	17.6	13.2	8.3	3.2	0.9
		MCWB	-0.8	-0.2	0.1	2.6	8.4	13.2	15.9	15.3	11.7	7.3	2.5	0.3
	10%	DB	-1.0	-0.5	0.5	3.1	9.6	14.9	17.8	16.1	12.2	7.3	2.5	0.4
		MCWB	-1.4	-1.0	-0.3	1.7	7.2	11.8	15.2	14.2	11.0	6.5	1.9	-0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	1.0	1.6	5.9	12.0	17.4	18.6	18.4	14.0	9.9	4.9	1.6
		MCDB	0.8	1.2	2.7	9.4	14.6	21.4	21.6	19.8	15.6	10.3	5.2	2.0
	2%	WB	0.0	0.4	0.7	3.9	10.1	15.0	17.4	16.6	12.7	8.7	4.0	0.9
		MCDB	0.2	0.7	1.3	5.8	13.0	18.2	19.9	18.6	13.7	9.3	4.7	1.2
	5%	WB	-0.7	-0.1	0.2	2.8	8.6	13.4	16.4	15.5	11.9	7.5	2.7	0.5
		MCDB	-0.5	0.2	0.9	4.2	11.3	16.2	18.7	17.3	13.0	8.2	3.2	0.8
	10%	WB	-1.4	-1.0	-0.2	2.0	7.2	12.0	15.3	14.4	11.2	6.6	1.9	0.0
		MCDB	-1.1	-0.6	0.4	2.9	9.6	14.7	17.4	15.9	12.1	7.2	2.4	0.3
Mean Daily Temperature Range	5% DB	MDBR	4.1	4.1	4.2	4.3	4.8	5.2	4.7	4.0	2.9	2.1	2.6	2.8
		MCDBR	4.0	3.9	3.4	5.4	8.3	8.8	7.3	6.0	4.2	2.4	2.6	2.6
		MCWBR	4.2	3.9	2.9	3.9	5.6	5.7	4.6	4.0	3.1	2.4	2.7	2.7
	5% WB	MCDBR	4.1	3.9	3.2	4.8	7.9	8.3	6.7	5.5	4.0	2.3	2.6	2.7
		MCWBR	4.1	3.9	2.8	3.7	5.8	5.8	4.7	4.2	3.2	2.3	2.8	2.8
Clear-Sky Solar Irradiance	taub	0.223	0.279	0.307	0.339	0.351	0.346	0.378	0.355	0.332	0.305	0.349	0.360	
	taud	1.953	2.069	2.116	2.226	2.432	2.509	2.448	2.488	2.527	2.353	2.543	2.581	
	Ebn at Noon	259	581	751	823	856	870	824	811	752	577	255	97	
	Edh at Noon	23	61	93	107	97	92	95	83	65	47	21	11	
All-Sky Solar Radiation	RadAvg	0.09	0.61	2.14	3.83	4.88	5.79	5.24	3.67	2.02	0.66	0.13	0.03	
	RadStd	0.01	0.13	0.24	0.34	0.38	0.54	0.50	0.40	0.20	0.10	0.02	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page